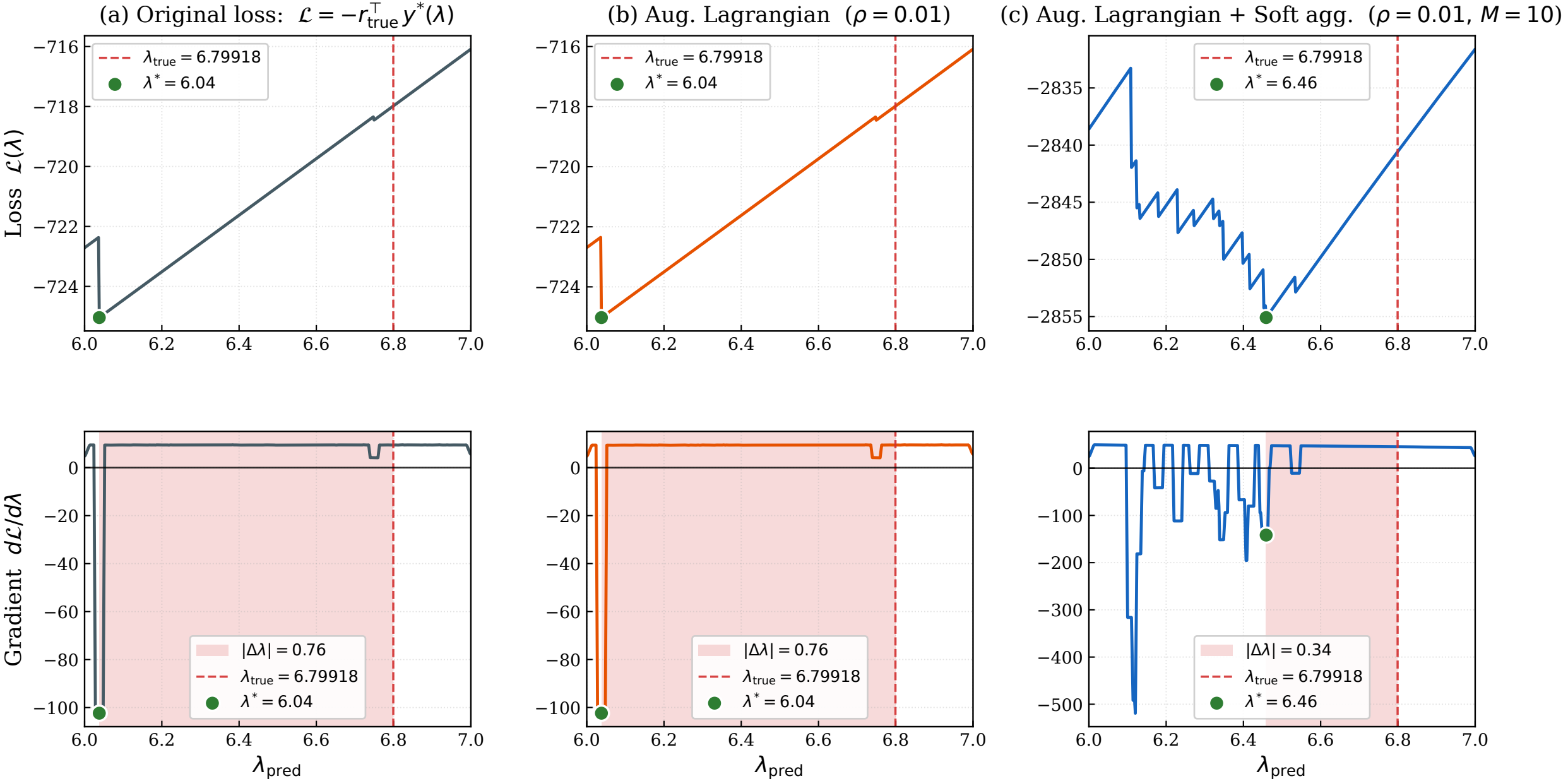
(a) Original loss: $\mathcal{L} = -r_{\text{true}}^{\top} y^*(\lambda)$
(b) Aug. Lagrangian ($\rho = 0.01$)
(c) Aug. Lagrangian + Soft agg. ($\rho = 0.01$, $M = 10$)
Loss $\mathcal{L}(\lambda)$
Gradient $d\mathcal{L}/d\lambda$
λ_{pred}
$\lambda_{\text{true}} = 6.79918$
$\lambda^* = 6.04$
$\lambda^* = 6.46$
$|\Delta\lambda| = 0.76$
$|\Delta\lambda| = 0.34$